

Dilution

David Frodsham

Diego Cioban

Venture investors invest in equity—shares or stock in a company—because they are taking an equity risk: if the company performs well, they want to benefit by owning a percentage share that might grow significantly in value, rather than being linked to the amount they have invested.

Founders start with 100% ownership of their company and become progressively diluted through rounds of capital increases. The mechanism is for the shareholders to authorize an increase in the share capital and for the company to issue enough new shares to meet its financial needs. If, for example, company Y with 100 shares, all owned by the founder, sells 33 new shares to an investor, the total number of shares in issue increases to 133.

After the round closes, the investor owns 33/133, approximately 25% of the company.

Most companies go through several rounds of investment before the “exit” or liquidity event—the point when the investors and the founders can convert some or all their shareholdings into cash.

Let us take the example of the above company Y, which raised a further three rounds of capital after the Series Seed round: the Series A, B, and C rounds, each with the same dilution of 25%. The percentage shareholding would look like that shown in **Table 1**.

Table 1
Evolution of shareholding in company Y

	Startup	Series Seed	Series A	Series B	Series C
Founders	100%	75%	56%	42%	32%
Seed investors		25%	19%	14%	11%
Series A investors			25%	19%	14%
Series B investors				25%	19%
Series C investors					25%

This technical note was prepared by David Frodsham, lecturer, and Diego Cioban, case writer. October 2023.

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The number of shares would evolve as shown in **Table 2**.

Table 2

Evolution of number of shares in company Y

	Startup	Series Seed	Series A	Series B	Series C
Founders	100	100	100	100	100
Seed investors		33	33	33	33
Series A investors			44	44	44
Series B investors				59	59
Series C investors					79
Total shares	100	133	178	237	316

The number of shares issued for each round increased to maintain the 25% dilution in each round.

Valuations are expressed as either pre- or post-money. It is easier to calculate the valuation by starting with the post-money, which is the amount invested divided by the percentage of the company owned by incoming investors. In company Y, the seed investor invested €1m for 25% ownership, representing a post-money valuation of €4m, being €1m divided by 25%.

The pre-money valuation is therefore €3m: the post-money valuation minus the amount invested. The price per share was the same in both cases: the investor bought 33 shares for €1m, or €30,000 per share. The valuations were as follows:

Pre-investment: 100 shares * €30,000 per share = €3m

Post-investment: 133 shares * €30,000 per share = €4m

Thus, the post-money valuation becomes the pre-money valuation in the subsequent round. The overall number of shares increases in each round to allow space for new investors, and the number of shares issued for each round increases as each round represents the same 25% dilution. The amount raised in each round also increases to match the increased valuation, as shown in **Table 3**.

Table 3

Evolution of valuation in company Y

	Startup	Series Seed	Series A	Series B	Series C
Amount raised		€1,000,000	€1,333,333	€1,777,778	€2,370,370
Price/share		€30,000	€30,000	€30,000	€30,000
Pre-money		€3,000,000	€4,000,000	€5,333,333	€7,111,111
Post-money		€4,000,000	€5,333,333	€7,111,111	€9,481,481

However, as the price per share in the given example is the same in every round, the valuation expansion is illusory.

Entrepreneurs are laser focused on the valuation of their ventures, but the correct focus should be on the share price expansion in each round. Institutional investors, on the other hand, focus on percentage ownership. They model an exit, make assumptions about the exit valuation and all the investment rounds needed, and work back to the holding they need at present.

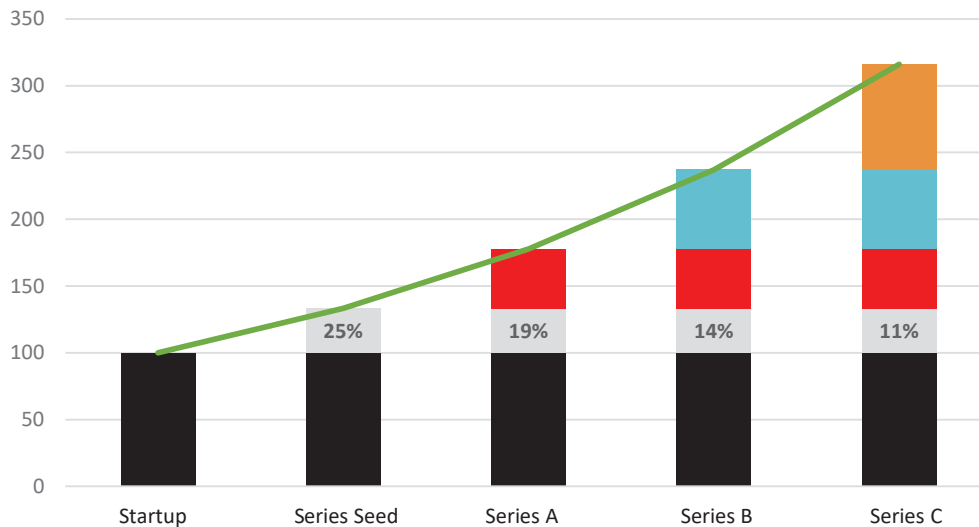


If, for example, an investor is targeting a €25m exit for their fund and expects the company to be worth €100m at that time, they need a percentage ownership of at least 25% by the time of the liquidity event. It follows that the investor should target a percentage ownership of at least 25% at the time of their first investment and seek to maintain that percentage by investing pro-rata in future rounds.

As a general point, it's problematic to increase percentage shareholdings in future rounds, because it would mean taking more than their pro-rata share of the round.. Thus, investors seek a percentage ownership that is at least as large as their exit target at the time of their initial investment.

Investors also need a mechanism to avoid dilution in subsequent rounds and to maintain their percentage shareholding. An example of the evolution of the seed investor's shareholding is given in **Figure 1**.

Figure 1
Number of shares by investment round



The mechanism in the term sheet for enabling pro rata investment rights is a preemption provision, which grants all shareholders the right to invest in each capital raise pro rata to their holding. In the example in **Figure 1**, the seed investor would be entitled to subscribe for up to 25% in the Series A round, for instance. Investors earmark (i.e., provisionally reserve) funds to invest in future rounds so that they can maintain their percentage ownership.

Of course, if all shareholders take up their full rights, there will be no room for a new investor. However, founders and seed investors often do not have enough capital to continue to invest in each round; seed investors acquire shares at a very low price and accept dilution as part of their investment model. For example, Pitchbook, a research firm that specializes in early-stage venture investing, reports that WhatsApp raised its Seed Round from angel investors at \$0.36 per share,



followed by institutional rounds at much higher prices of \$3.79 per share (Series A round) and \$6.53 (Series B round).

The foregoing squeezes the availability of space in the cap table, leading to either less room for future investors or greater amounts that need to be raised, which may further dilute the founders and seed investors. **Table 4** shows how the cap table would look if the institutional investors maintained their shareholdings and the later-round investors accepted a smaller percentage holding.

Table 4
Effect of the exercise of preemption rights on later investors

	Startup	Series Seed	Series A	Series B	Series C
Founders	100%	75%	56%	42%	32%
Series Seed		25%	19%	14%	11%
Series A			25%	25%	25%
Series B				19%	19%
Series C					14%

The overall number of shares, the pre- and post-money valuations, and the percentages owned by the founders and seed investors remain the same. If that is not acceptable to Series B and later investors because it does not allow them to achieve their ownership objectives, more shares will be issued at the same price per share, the post-money valuation will increase, and there will be more dilution for the founders. It is quite common to hear of rounds being “extended” on the same terms, which is what this represents.

As the cap table gets filled with investors guarding their percentage ownership, it becomes increasingly difficult to bring in a new investor at a high percentage ownership, and it is common for later shareholders to own smaller slices of the company. This is understandable when risk is considered. Early-stage investors take risks on everything—team, technology, business model, and market—and require a high potential rate of return to compensate for their portfolio losses and underperformance. With each round of investment, risks are reduced because the technology employed works, a team is in place, there are customers, and other commercial proof points. Thus, later-stage investors can be successful, but with a lower target return.

In fact, albeit slightly off-topic for this paper, each round of investment is often timed to be around risk reduction, such as proof of concept, first customer, patents applied for, or partners in place. Founders may justify a funding round and a valuation increase on the basis that development of a certain technology has been completed, or that first sales have been made to their target market, or that sales are growing strongly; and additional funding is required to move to full production, build a sales team, or scale the platform.

The aforementioned example assumes that the price per share remains the same, or that although the valuation is increasing, the increase is not enough to reward the earlier shareholders for having taken more risks by committing funds earlier. In the example in **Table 3** the price per share remains the same from one investment round to the next – although valuation is increasing, the company is not progressing in valuation enough to warrant a higher per-share price.



Let us assume that there has, in fact, been an appropriate risk reduction and that the price per share increases by 20% from round to round. The cap table may not change in terms of percentage ownership; rather what changes is the valuation in each round and the amount raised (see Table 5).

Table 5

The effect of share price increases on the valuations and amounts raised

	Seed	Series A	Series B	Series C	Total
Amount raised	€1,000,000	€1,666,667	€2,777,778	€3,703,704	€9,148,148
Price/share	€30,000	€37,500	€46,875	€58,594	
Pre-money	€3,000,000	€4,000,000	€6,666,667	€11,111,111	
Post-money	€4,000,000	€6,666,667	€11,111,111	€18,518,519	

For the same dilution, the company has raised over €9m instead of €6m. For a total investment of about €2.8m, the Series A investors have acquired the following:

- 44 shares at €30,000 = €1,333,333
- 15 shares at €37,500 = €555,556
- 20 shares at €58,594 = €925,926

In the aforementioned example, if the reasonable expectation is for the company to achieve a valuation of €100m at exit, and if the Series A investor is targeting a 10x return, the model implies that their €2.8m investment should be worth €25m at exit, quite close to their target.

Professional investors thus keep two to three times their original investment reserved for future rounds. As each portfolio company investment is commonly capped at 15% of the size of the fund, the first investment should not exceed 5% of the fund.

The investor Investment Committee will expect to see an analysis of the total expected return to the fund in all investment rounds. To estimate this, the team presenting the investment opportunity to the investment committee and seeking its approval will need to estimate the following:

- Exit valuation of the business. This can be done by using multiples (usually revenue or profits) at comparable companies and adjusting for different growth rates (higher = higher valuation), market share (higher = higher valuation), or geography (companies in the United States, for example, are generally valued higher than those in Europe).
- Total capital required up to exit. There are two ways to consider this:
 - Identify and quantify the drivers of capital need (product, platform, and other investments; working capital; unit economics; and cumulative losses to breakeven).
 - Map the milestones that define the capital rounds. This helps establish the number of rounds needed and the capital that needs to be raised at each point.



The following is an example of what the investment committee will expect to hear:

We have identified company Y as an interesting investment target (for reasons a, b, and c). We are seeking approval to invest €1 million in the Series A round for 25% of the capital. We believe that the total capital requirement will be approximately €10m in this and the next two rounds, and that we will require a total of €2.5m to €3m to maintain our 25% holding. We expect the exit valuations to be similar to these for [at least two] companies, implying a reasonable exit target of €100 million. The multiple return for our fund will therefore be about 10x.

Founders' shares and Employee Stock Ownership Plan (ESOP)

Founders' shares—issued at no cost or at a low cost to the founders at the time of a company's creation—and employee share option plans (ESOP)—the option to acquire shares at a future date at a defined price—are commonly subject to vesting (i.e. the holders become owners over time) to encourage retention and to enable shares to be clawed back in the event that an employee leaves.

For the purposes of the cap table, vesting is ignored, and the full amount of the founders' shares as well as the full amount reserved for the ESOP—granted or not—is shown in the table. This means that at the time of an exit, there may be some accretion to shareholders as ungranted options and unvested ESOP options and founders' shares are excluded.

It is common for investors to top up the ESOP at the time of each round, to reduce dilution for existing option holders, and to create space in the ESOP for the recruitment of new key employees as the company grows and expands.

Convertible loan note (CLN) and Simple Agreement for Future Equity (SAFE)

As mentioned above, VC investors invest in equity¹ because they are taking an equity risk. CLNs provided by investors are converted into equity later and are a mechanism for having a preferred investment position in a negative scenario and an equity return if the performance is good.

CLNs are used to ensure that the company does not run out of money in the run-up to the next investment round. They are provided by existing shareholders, and as an incentive to them, CLNs are converted into shares at a defined discount (often 20%), plus an interest rate and perhaps a valuation cap.

Combinator, an incubator of startups, devised a CLN for very early-stage companies, called a SAFE, allowing micro-investment commitments to be made over a period of time, with the share price being defined later, depending on the demand from investors. Again, the investment should be considered equity.

CLNs and SAFE are not reflected in the cap table until they are converted into shares, although a footnote is often made on the cap table to the effect that there are outstanding CLNs.

¹ There are exceptions, such as venture debt, whose model is to accrue or charge a high percentage interest and enhance returns with share warrants, also a form of equity.



Economic ownership versus percentage ownership

Percentage ownership rarely equates exactly to the percentage of proceeds on exit. The main reason for this is the presence of liquidation preferences in investor shares, but it can also be due to participation, accrued dividends, anti-dilution, CLN discounts, warrants, ESOPs and other investment terms. This is further complicated by the fact that exits often include payment in shares, cash tranches, earn-out provisions, and escrow holdbacks for warranty claims.

Working out who gets what in the event of a sale is called a waterfall and is beyond the scope of this paper, other than to say that it easily becomes very complicated and is important to model in the run-up to an exit process.